

Optimal Market Making for Cryptocurrency Options

The Guéant-Lehalle-Fernandez-Tapia Framework and its Extensions

A 10-Week Summer Research Program

North Carolina State University

Industry Mentor

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San Francisco

2026

1. Course Overview

This ten-week summer research program prepares students in the North Carolina State University Financial Mathematics graduate program to design, implement, and evaluate optimal market making strategies for cryptocurrency options. The program is organized around the Guéant–Lehalle–Fernandez-Tapia (GLFT) framework, a modern stochastic-control approach to liquidity provision that generalizes the earlier work of Avellaneda and Stoikov. The central pedagogical question is how to extend GLFT, which was originally formulated for linear instruments, to non-linear derivatives whose inventory risk is multi-dimensional in the Greeks.

The program is mentored by Dendi Suhubdy. Students will work with live and historical Coincall data throughout the program and will produce a research-grade final report and code repository suitable for further academic or industry use.

1.1 Learning Objectives

Upon successful completion of this program, students will be able to:

1. Formulate a market making problem as a continuous-time stochastic optimal control problem and derive the associated Hamilton–Jacobi–Bellman (HJB) equation.
2. Solve the HJB system for the Avellaneda–Stoikov and Guéant–Lehalle–Fernandez-Tapia models, including the asymptotic regime in which closed-form quote expressions are available.
3. Calibrate parametric volatility surfaces (SVI, SABR) to live cryptocurrency options data and assess their stability across time.
4. Implement production-grade fast pricing routines using characteristic function methods (Carr–Madan FFT, Fang–Oosterlee COS) and compute the full Greek vector via adjoint algorithmic differentiation.
5. Extend the GLFT framework to a multi-Greek inventory penalty following El Aoud and Abergel, and reduce the resulting HJB system to a numerically tractable form.
6. Integrate a dynamic hedging policy with the quoting engine and analyze the resulting P&L attribution.
7. Build and validate an event-driven backtest that consumes real Coincall order book data and produces interpretable P&L, risk, and Greek-exposure diagnostics.
8. Communicate research results in writing and orally at a level suitable for industry quantitative research interviews.

1.2 Prerequisites

Students are expected to have completed or be concurrently enrolled in graduate coursework covering: stochastic calculus (Itô integration, Itô's lemma, Girsanov's theorem), continuous-time stochastic processes (Brownian motion, Poisson processes, jump-diffusions), Black–Scholes–Merton option pricing, partial differential equations at the level of Evans, and probability theory at the level of Durrett. Programming maturity in Python is required; familiarity with NumPy, SciPy, and JAX is advantageous but not required.

1.3 Structure

Each week of the program comprises two ninety-minute lectures, one ninety-minute recitation, and approximately fifteen hours of independent reading, problem-solving, and programming. Weeks 5, 7, and 9 are designated milestone weeks at which students deliver a working software component built on top of the preceding weeks' material. The final week is devoted to a chosen extension topic and the preparation and presentation of a research-grade final report.

The compression from a longer original schedule to ten weeks places additional self-study expectations on students: in particular, the survey of GLFT variants (multi-asset, adverse selection, discrete-inventory) is moved from a dedicated lecture week into required independent reading during week 5, and the practitioner-oriented treatment of dynamic hedging is integrated into the options market making theory week rather than receiving a standalone week. Students are expected to complete the corresponding readings on schedule.

1.4 Assessment

Final grades are determined as follows. Weekly problem sets account for thirty percent of the grade. The three milestone deliverables in weeks 5, 7, and 9 jointly account for thirty percent. The final report and presentation account for thirty percent. Active participation in lecture, recitation, and code review accounts for the remaining ten percent.

1.5 Mentorship Model

Students will have access to weekly thirty-minute one-on-one office hours with the industry mentor, in addition to scheduled lectures and recitations. Office hours are intended for clarification of difficult material, code review of milestone deliverables, and discussion of research directions for the final week. Students are expected to come to office hours with specific questions and prepared work.

2. Program Milestones at a Glance

The program is organized around three milestone deliverables. Each milestone integrates the material from the preceding weeks and produces a software artifact that is incorporated into the final research repository.

Week	Title	Deliverable
5	Linear-asset GLFT engine	Working Python implementation of the GLFT model on a single linear asset, validated against simulated dynamics and benchmarked on one month of Coincall BTC-PERP order book data.
7	Fast pricing and Greeks library	Benchmarked pricing and Greeks library for European options under Black-Scholes and Heston dynamics, with characteristic function pricing (FFT and COS) and full Greek vector computation via algorithmic differentiation. Latency target: full Coincall BTC surface revaluation in under 5 milliseconds.
9	Full options market making backtest	Event-driven backtest of the multi-Greek market making engine on three months of Coincall BTC and ETH options data, with full P&L attribution, risk metrics, and Greek-exposure time series.

3. Week-by-Week Schedule

Each of the following ten sections gives the learning objectives, lecture outlines, recitation activity, required reading, problem set, and (where applicable) milestone deliverable for one week of the program. Sections are designed to be read end-to-end as a working syllabus for instructors and as a study plan for students.

Week 1 — Foundations of Market Microstructure

Theme: from limit order books to the economic role of the market maker.

Learning Objectives

- Reconstruct a limit order book from raw exchange feeds and compute standard liquidity metrics.
- Articulate the three classical components of the bid-ask spread and identify them in real data.
- Explain the role of the market maker as a continuously quoting liquidity provider, and contrast with directional traders.

Lecture 1 — Limit Order Book Mechanics

An introduction to the institutional setting in which the rest of the program takes place. The lecture begins with the mechanics of central limit order books on modern electronic exchanges, then introduces order types, matching rules, and the basic vocabulary of microstructure. The lecture closes with a tour of the Coincall exchange architecture as a concrete example.

- Order types: market, limit, IOC, post-only, hidden.
- Price-time priority and the matching algorithm.
- Order book aggregation: L1, L2, L3 views and what each reveals.
- Tick size, lot size, and the discrete grid of available prices.
- Coincall-specific architecture: contract specifications, fee structure, market maker tiers.

Lecture 2 — Spread Decomposition and the Role of the Market Maker

The bid-ask spread is the market maker's source of revenue, but it is also the compensation demanded for three distinct costs. This lecture decomposes the spread into adverse selection, inventory, and order-processing components, drawing on the classical models of Roll, Glosten–Milgrom, and Kyle.

- The Roll (1984) model: estimating effective spread from trade prices alone.
- Glosten–Milgrom (1985): asymmetric information and the rational quote-setter.
- Kyle (1985): the strategic informed trader and market depth.
- Empirical decomposition: realized spread, effective spread, price impact.
- Why crypto markets exhibit different microstructure characteristics than equities.

Recitation

Hands-on session reconstructing the order book from a Coincall tick-level recording. Students will load one hour of BTC-PERP feed data, replay the events in time order, and produce a time series of mid-prices, spreads, and book depth. Output: a Python class that maintains a live order book state from raw events.

Required Reading

- Cartea, Jaikumar, and Penalva (2015), *Algorithmic and High-Frequency Trading*, Chapters 1–3.
- Roll (1984), *A Simple Implicit Measure of the Effective Bid-Ask Spread*.
- Glosten and Milgrom (1985), *Bid, Ask and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders*.

Problem Set 1

1. Implement an order book reconstruction class in Python that consumes a stream of order events (new, cancel, modify, trade) and maintains the current state of the L2 book. Apply it to a one-hour sample of Coincall BTC-PERP data and produce a plot of the mid-price and quoted spread over time.
2. Compute the Roll estimator of the effective spread on the same data sample, and compare it to the directly observable bid-ask spread. Comment on any discrepancy.
3. Derive analytically the quote-setter's bid and ask in the Glosten–Milgrom model under the assumption of a single informed trader and a Bernoulli arrival of informed versus uninformed orders.
4. Numerically simulate the Glosten–Milgrom model with parameters of your choice and verify your derivation in problem 3. Plot the convergence of the public belief about asset value as trades arrive.
5. Compute the realized spread and price impact on one hour of Coincall data using a five-minute decomposition window. Comment on what these metrics imply about the toxicity of the order flow.

Week 2 — The Avellaneda–Stoikov Model

Theme: a fully worked example of optimal market making through stochastic control.

Learning Objectives

- Set up the Avellaneda–Stoikov market making problem as a continuous-time stochastic control problem with exponential utility.
- Derive the associated HJB equation and reduce it to a system of ODEs through the appropriate ansatz.
- Solve the system in closed form under the simplifying assumptions of the original paper and interpret the resulting quote formulas economically.

Lecture 1 — Setup and Dynamics

The lecture introduces the Avellaneda–Stoikov framework: a market maker quoting bid and ask prices on a single asset whose mid-price follows arithmetic Brownian motion, with order arrivals modeled as Poisson processes whose intensities depend exponentially on the quote distance from the mid. The wealth and inventory processes are derived, and the optimization problem is stated.

- Mid-price dynamics: arithmetic Brownian motion as a local approximation.
- Poisson order flow with intensity $\lambda(\delta) = A \exp(-\kappa\delta)$.
- Wealth and inventory state dynamics under continuous quoting.
- Exponential utility: motivation, properties, and why it makes the problem tractable.
- The terminal liquidation penalty and finite-horizon formulation.

Lecture 2 — HJB Derivation and Solution

Starting from the dynamic programming principle, the lecture derives the HJB equation for the value function and applies the standard ansatz that separates the value function into wealth, inventory, and time components. The resulting reduced system of ODEs admits an asymptotic closed-form solution that exposes the structure of the optimal quotes.

- The dynamic programming principle and the HJB equation.
- Separation ansatz: $V(t, x, S, q) = -\exp(-\gamma(x + qS)) \theta(t, q)$.
- Reduction to a system of ODEs in $\theta(t, q)$.
- The reservation price $r(s, q) = s - q\gamma\sigma^2(T - t)$.
- Optimal half-spread expression and its decomposition into a risk-aversion and an order-flow component.

Recitation

Workshop on the derivation. Students bring partial derivations to the recitation and complete them in pairs. The mentor walks through the verification theorem step by step. Output: a complete typed derivation in LaTeX, to be referenced throughout the program.

Required Reading

- Avellaneda and Stoikov (2008), High-Frequency Trading in a Limit Order Book.

- Cartea, Jaikumar, and Penalva (2015), Chapter 10.

Problem Set 2

1. Derive the HJB equation for the Avellaneda–Stoikov model from first principles, justifying each step. Show that the value function ansatz $V(t, x, S, q) = -\exp(-\gamma(x + qS)) \theta(t, q)$ reduces the HJB to a system of ODEs in θ .
2. Solve the reduced ODE system in the asymptotic regime $T \rightarrow \infty$ and obtain the closed-form expressions for the reservation price and optimal half-spread.
3. Implement the Avellaneda–Stoikov quote engine in Python. Simulate one trading day of mid-price dynamics with $\sigma = 2 \text{ USD}/\sqrt{s}$, fill arrivals with $\kappa = 1.5$ and $A = 140$, and compare the P&L distribution of the optimal strategy against a symmetric quoting strategy that ignores inventory.
4. Perform a sensitivity analysis on the risk-aversion parameter γ . Plot inventory variance and Sharpe ratio against γ over a range covering three orders of magnitude, and identify the regime where the model is most effective.
5. Discuss the assumption that mid-price dynamics are independent of the market maker's quotes. Identify at least three empirical phenomena in real markets that violate this assumption and discuss how each would affect the optimal strategy.

Week 3 — Stochastic Optimal Control in Continuous Time

Theme: the general theory underlying the previous week's worked example.

Learning Objectives

- State and apply the dynamic programming principle in a general continuous-time setting.
- Derive the HJB equation for a generic stochastic control problem and articulate the conditions of the verification theorem.
- Recognize the role of CARA and CRRA utilities in producing tractable problems and identify the consequences for the inventory penalty.

Lecture 1 — Dynamic Programming and the HJB Equation

The lecture develops the dynamic programming principle in continuous time and derives the HJB equation as a consequence. The treatment is rigorous but informal: viscosity solutions are mentioned as the natural setting for non-smooth value functions, but the program will focus on classical solutions throughout.

- Controlled diffusions and admissible controls.
- The dynamic programming principle: heuristic and formal statement.
- Infinitesimal generator of a controlled diffusion.
- Derivation of the HJB equation by passing to the limit.
- Boundary and terminal conditions; finite vs infinite horizon.

Lecture 2 — Verification Theorems and Tractable Utility Functions

The HJB equation is a necessary condition; the verification theorem provides the converse, stating that a sufficiently smooth solution of HJB is the value function. The lecture then discusses why exponential (CARA) utility produces tractable problems in market making, and why power (CRRA) utility appears in consumption-investment problems instead.

- Statement and proof sketch of the verification theorem.
- Smooth-fit conditions at free boundaries.
- Viscosity solutions: when classical solutions fail.
- CARA, CRRA, and HARA utility families; properties and use cases.
- Why CARA decouples the value function from wealth in market making.

Recitation

Worked solutions to classical control problems: Merton's portfolio problem under CARA and CRRA utility, an optimal liquidation problem with linear price impact. Students will identify the pattern of HJB derivation, ansatz selection, ODE reduction, and solution that will recur throughout the program.

Required Reading

- Pham (2009), Continuous-time Stochastic Control and Optimization with Financial Applications, Chapters 3–4.
- Cartea, Jaikumar, and Penalva (2015), Chapter 6.

Problem Set 3

1. Solve the Merton portfolio problem in continuous time under CRRA utility. Derive the HJB equation, propose an ansatz, reduce to an ODE, and solve for the optimal consumption and investment policy.
2. Repeat the previous problem under CARA utility and contrast the structure of the solution. Explain economically why the optimal portfolio depends on wealth under CRRA but not under CARA.
3. Solve the Almgren–Chriss optimal execution problem in its continuous-time formulation. Derive the optimal trading trajectory under quadratic transient impact and verify your solution against the original discrete-time result.
4. State the verification theorem in your own words and identify the precise role of each assumption (smoothness, growth conditions, admissibility class). Construct a counterexample in which one assumption fails and the conclusion of the theorem fails as well.
5. Read the introduction of Pham Chapter 4 on viscosity solutions and explain in two paragraphs why viscosity solutions are needed in problems with non-smooth value functions, using the obstacle problem as your example.

Week 4 — The GLFT Model, Part I — Setup and HJB Derivation

Theme: generalizing Avellaneda–Stoikov to arbitrary intensity functions.

Learning Objectives

- State the GLFT problem setup and identify the assumptions that distinguish it from Avellaneda–Stoikov.
- Derive the HJB system for the GLFT problem and recognize the role of the intensity functions Λ^+ and Λ^- .
- Recover the Avellaneda–Stoikov solution as a special case of GLFT with exponential intensities and verify the limiting behavior.

Lecture 1 — The GLFT Setup

The lecture introduces the Guéant–Lehalle–Fernandez-Tapia framework. The setup is similar to Avellaneda–Stoikov but generalizes in two important directions: arbitrary intensity functions (subject to regularity conditions), and a clean treatment of an inventory constraint that bounds the position from below and above.

- Generalized intensity functions $\Lambda^+(\delta)$ and $\Lambda^-(\delta)$: regularity assumptions.
- Inventory constraint and the truncation of admissible quotes at the boundary.
- Wealth and cash dynamics; the cash process as a state variable.
- Comparison to Avellaneda–Stoikov: what generalizes, what does not.
- Discussion of the asymptotic regime that will become central next week.

Lecture 2 — HJB Derivation in the GLFT Framework

The lecture derives the HJB system carefully, paying attention to the boundary behavior at the inventory limits. The same separation ansatz used in Avellaneda–Stoikov is applied, but the resulting reduced system is a coupled system of ODEs that does not admit a closed-form solution for general intensities.

- HJB equation in the multi-jump setting.
- The separation ansatz $V(t, x, S, q) = -\exp(-\gamma(x + qS)) \theta(t, q)$ and its validity.
- Reduced ODE system in $\theta(t, q)$: coupling structure across q .
- Boundary conditions at the inventory limits $q_{\min}$ and $q_{\max}$.
- Verification of the Avellaneda–Stoikov limit when $\Lambda(\delta) = A \exp(-\kappa\delta)$.

Recitation

Pair derivation session. Students re-derive the GLFT HJB step by step, with the mentor checking work in real time. The recitation closes with a discussion of how the choice of intensity function reflects assumptions about the order flow that the market maker faces.

Required Reading

- Guéant, Lehalle, and Fernandez-Tapia (2013), Sections 1–3.
- Guéant (2016), The Financial Mathematics of Market Liquidity, Chapter 9.

Problem Set 4

1. Derive the HJB system for the GLFT problem from first principles, carefully justifying the treatment of the inventory boundary. Identify the precise step at which the regularity assumptions on Λ are needed.
2. Show analytically that the GLFT HJB system reduces to the Avellaneda–Stoikov HJB when $\Lambda(\delta) = A \exp(-\kappa\delta)$ and the inventory constraint is relaxed.
3. Consider three families of intensity functions: exponential, power-law, and logistic. For each, plot the implied order arrival rate as a function of quote distance and discuss what assumption about trader behavior the family encodes.
4. Derive the inventory dynamics under a generic intensity function Λ and compute the expected change in inventory over a short interval, conditional on the current inventory and the chosen quotes.
5. Read Section 2 of Guéant, Lehalle, and Fernandez-Tapia (2013) and write a one-page summary of the modeling choices that differ from Avellaneda–Stoikov. Comment on which differences are mathematically substantive and which are notational.

Week 5 — The GLFT Model, Part II — Asymptotic Solution, Implementation, and Variants

Theme: the asymptotic regime in which GLFT admits closed-form quotes, a working implementation, and a self-study survey of the GLFT variants literature.

Learning Objectives

- Derive the closed-form GLFT quote expressions in the asymptotic regime via the appropriate change of variable.
- Implement the GLFT engine in Python, validated against simulated dynamics.
- Backtest the engine against Coincall BTC-PERP order book data and produce P&L, inventory, and Sharpe diagnostics.
- Read and summarize the major extensions of GLFT (multi-asset, predictive signals, adverse selection, discrete inventory) for use in the final-week extension project.

Lecture 1 — The Asymptotic Regime and Closed-Form Quotes

The technical centerpiece of the GLFT paper. Under the asymptotic regime in which the trading horizon is far from the terminal date, the change of variable $u(q) = \theta(t, q)/\theta(t, q+1)$ linearizes the reduced ODE system into an eigenvalue problem, whose dominant eigenvector encodes the optimal quoting policy. The lecture works through this derivation carefully. The lecture closes with a brief survey of variants (multi-asset, predictive signals, adverse selection, discrete inventory) that students will read about independently this week as preparation for the final-week extension topic.

- The asymptotic regime: motivation and conditions.
- Change of variable $u(q)$ and the resulting linear system.
- Eigenvalue problem and the role of the Perron–Frobenius theorem.
- Closed-form expressions for the optimal bid and ask quotes.
- Decomposition of the quote into a 'fair price adjustment' and a 'spread' component.
- Brief survey: pointers to multi-asset, signal, and adverse-selection extensions in Guéant (2016).

Lecture 2 — Numerical Implementation and Validation

The lecture transitions from theory to code. The closed-form quotes are computed via solution of a linear system; the simulation infrastructure is built from the order book and trade-arrival models studied in week 1. The bulk of the lecture is dedicated to validation: comparing simulated P&L distributions against theoretical predictions, and identifying common implementation pitfalls.

- Solving the linear system for $u(q)$ numerically with sparse linear algebra.
- Mapping the model's continuous time to discrete event arrivals in simulation.
- Calibrating intensity functions Λ^+ , Λ^- from historical Coincall fill data.
- Backtest harness: replay engine, fill simulator, P&L attribution.
- Common bugs: off-by-one inventory indexing, sign errors in cash dynamics, miscalibrated intensities.

Recitation

Code review of student implementations. Each student presents their implementation and walks through one complete simulated trading day. The mentor identifies issues in real time and the group resolves them collectively. Time permitting, the recitation closes with a brief discussion of the variants reading.

Required Reading

- Guéant, Lehalle, and Fernandez-Tapia (2013), Sections 4–6.
- Guéant (2016), Chapters 9–11 (independent self-study; required as the survey of GLFT variants in lieu of a dedicated lecture week).

Problem Set 5

1. Carry out the change of variable $u(q) = \theta(t, q)/\theta(t, q+1)$ in the GLFT reduced ODE system and show that, in the asymptotic regime, the result is a linear system whose solution determines the optimal quotes.
2. Solve the linear system numerically for a problem with inventory limits $q \in \{-10, \dots, +10\}$ and intensity functions $\Lambda^+(\delta) = \Lambda^-(\delta) = 1.0 \exp(-1.5\delta)$. Plot the optimal bid and ask half-spreads as functions of inventory.
3. Reproduce Figure 2 of Guéant, Lehalle, and Fernandez-Tapia (2013) on simulated data with the parameters given in the paper. Comment on any discrepancies.
4. Calibrate the intensity functions from one week of Coincall BTC-PERP fill data using maximum likelihood. Report the fitted parameters and goodness-of-fit diagnostics.
5. Run a complete one-month backtest of your GLFT implementation against Coincall BTC-PERP data and report mean P&L, P&L volatility, Sharpe ratio, maximum drawdown, mean inventory, and inventory standard deviation. Compare against a baseline that quotes symmetrically without inventory adjustment.
6. Submit a two-page summary of Guéant (2016) Chapters 10–11, identifying one extension you find most relevant to crypto options market making and articulating why. This summary will be returned in week 10 as a candidate extension topic.

Milestone Deliverable: Milestone 1 — Linear-Asset GLFT Engine

Submit a Python package implementing the GLFT optimal market making model on a single linear asset, together with calibration, simulation, and backtesting infrastructure. The deliverable must include a reproducible backtest on one month of Coincall BTC-PERP data, with results documented in a short technical report following the structure of the original GLFT paper.

Datasets provided by Coincall:

Dataset	Contents and Coverage
coincall_btc_perp_l2_2025Q4.parquet	BTC-USDT perpetual futures L2 order book snapshots at 100ms resolution for October–December 2025. Ten levels each side. Columns: timestamp, bid_px_0..9, bid_sz_0..9, ask_px_0..9, ask_sz_0..9, sequence_number.

coincall_btc_perp_trades_2025Q4.parquet	BTC-USDT perpetual futures trade tape for the same period. Columns: timestamp, price, size, aggressor_side, trade_id.
coincall_intensity_calibration_set.parquet	Pre-aggregated fill events grouped by quote distance from mid, suitable for maximum likelihood calibration of intensity functions. Provided as a convenience; students are expected to also produce the same aggregation from the raw L2 and trade data above.

Acceptance criteria:

- GLFT quote computation runs in under 100 microseconds per update on a single CPU core.
- Intensity calibration produces a non-trivial fit, with χ^2 goodness-of-fit reported.
- Backtest produces P&L within $\pm 50\%$ of a reference implementation provided by the mentor.
- Code passes linting and includes unit tests covering the core quote computation.
- Technical report is at most ten pages excluding appendices.

Week 6 — Options Pricing, the Volatility Surface, and Calibration

Theme: a single concentrated week on options theory and the practical construction of a volatility surface.

Learning Objectives

- Recall the Greeks of European options and their economic interpretation, including the second-order Greeks vanna and volga.
- State the no-static-arbitrage constraints on the volatility surface and verify them on parametric surfaces.
- Calibrate SVI and SABR parameterizations to live Coincall options data with attention to stability.

Lecture 1 — Greeks and the Volatility Surface

The lecture begins with a brisk review of European option pricing under Black-Scholes and the Greeks delta, gamma, vega, theta, vanna, and volga. The empirical phenomenon of the volatility smile and the term structure of implied volatility is then introduced, along with Gatheral's formulation of the no-static-arbitrage conditions. The lecture concludes with a discussion of the specific features of the cryptocurrency options surface.

- Black-Scholes formula and the Greek vector: closed forms and identities.
- Vanna, volga, and their role in volatility risk.
- The implied volatility smile and term structure: empirical regularities.
- Calendar arbitrage, butterfly arbitrage, and the corresponding inequalities.
- Crypto options surfaces: sparse strikes, wide wings, and quote staleness.

Lecture 2 — Parameterization and Calibration

The lecture covers the two dominant parametric families: SVI (stochastic volatility inspired, Gatheral) and SABR (stochastic alpha-beta-rho, Hagan et al.). Each is introduced, its no-arbitrage form discussed, and its calibration procedure derived. The lecture spends substantial time on the practical issues of calibration to noisy data.

- Raw SVI parameterization and the arbitrage-free 'eSSVI' formulation.
- Calibration of SVI by nonlinear least squares; sensitivity to initial guess.
- SABR formula and the small-time expansion of Hagan et al.
- Time-stability of calibrated parameters: smoothing across days.
- Diagnostic plots: residuals, leave-one-out errors, parameter time series.

Recitation

Live calibration session. Students load a Coincall options chain snapshot, fit SVI and SABR surfaces, and produce the standard diagnostic plots. The session ends with each student comparing their calibration to the others and discussing the sources of variation.

Required Reading

- Gatheral (2006), *The Volatility Surface*, Chapters 1–5.
- Gatheral and Jacquier (2014), *Arbitrage-Free SVI Volatility Surfaces*.

- Hagan, Kumar, Lesniewski, and Woodward (2002), Managing Smile Risk.
- Hull (2021), Chapters 15, 19–21 (review of Greeks and basic surface concepts).

Problem Set 6

1. Derive the closed-form expressions for vanna and volga under Black–Scholes. Verify your derivation by finite-difference computation against the Black–Scholes price and report the maximum relative error across a range of moneyness and maturities.
2. Implement the raw SVI parameterization and the arbitrage-free constraints from Gatheral and Jacquier. Apply the constraints to a deliberately misparameterized surface and observe which constraints are violated.
3. Calibrate SVI to one Coincall BTC options chain snapshot using nonlinear least squares. Report the fitted parameters, residuals at each strike, and the resulting RMS error in implied volatility units.
4. Calibrate the SABR model to the same snapshot. Compare its fit quality against SVI and discuss the strengths and weaknesses of each parameterization for crypto options.
5. Run daily SVI calibration over one month of Coincall data and produce a time series of each SVI parameter. Comment on the stability of the calibration and suggest a smoothing scheme that would improve day-to-day consistency.

Week 7 — Fast Computation of Prices, Greeks, and Surfaces

Theme: the latency budget of an options market maker, and the numerical methods that meet it.

Learning Objectives

- Implement and benchmark characteristic function pricing methods (Carr–Madan FFT and Fang–Oosterlee COS).
- Compute the full Greek vector via adjoint algorithmic differentiation rather than finite differences.
- Vectorize pricing and Greek computation across an entire surface to meet a production latency budget.

Lecture 1 — Characteristic Function Pricing

Closed-form Black–Scholes pricing covers only one model. As soon as students consider Heston or any stochastic-volatility model for the underlying, they need numerical pricing infrastructure. The lecture covers the two dominant approaches in the practitioner literature: the Carr–Madan FFT method, which exploits the structure of the call price under exponential damping, and the Fang–Oosterlee COS method, which uses Fourier-cosine series and converges geometrically for smooth payoffs.

- The characteristic function of log-returns under Black–Scholes and Heston.
- The Carr–Madan FFT method: dampened payoff, FFT, recovery of prices.
- The Fang–Oosterlee COS method: truncation interval, cosine coefficients, convergence.
- Comparison: when each method dominates in accuracy or speed.
- Monte Carlo as the baseline; variance reduction by control variates.

Lecture 2 — Algorithmic Differentiation and Vectorization

Finite-difference Greeks are correct but slow: each Greek requires re-evaluating the price, and a market maker quoting on dozens of contracts with several Greeks per contract quickly exhausts the latency budget. Adjoint algorithmic differentiation (AAD) computes the entire Greek vector in time comparable to a single price evaluation. The lecture covers forward-mode AD, reverse-mode AD, and their use in derivatives pricing, with concrete examples in JAX and a brief discussion of C++ implementations via Adept and Enzyme.

- Forward-mode and reverse-mode automatic differentiation: theory and complexity.
- Smoking adjoints: Giles and Glasserman's foundational application to Monte Carlo Greeks.
- JAX in practice: `grad`, `jacfwd`, `jacrev`, `vmap`, `jit`.
- Pathwise versus likelihood-ratio Greeks: when each is applicable.
- Vectorization across a surface: SIMD, JAX `vmap`, and GPU offload.
- C++ AAD: Adept, Stan Math, and the emerging compiler-level option via Enzyme.

Recitation

Hands-on optimization session. Each student profiles a naive implementation of surface revaluation, identifies the bottleneck, applies vectorization and AAD, and re-measures. The goal is a measured speedup of at least one order of magnitude over the naive baseline.

Required Reading

- Carr and Madan (1999), Option Valuation Using the Fast Fourier Transform.
- Fang and Oosterlee (2008), A Novel Pricing Method for European Options Based on Fourier-Cosine Series Expansions.
- Giles and Glasserman (2006), Smoking Adjoints: Fast Monte Carlo Greeks.
- Capriotti (2011), Fast Greeks by Algorithmic Differentiation.
- Glasserman (2003), Monte Carlo Methods in Financial Engineering, selected sections on variance reduction.

Problem Set 7

1. Implement the Carr–Madan FFT pricer for European calls under the Heston model. Verify against a Monte Carlo benchmark and report the convergence rate as a function of the FFT grid size.
2. Implement the Fang–Oosterlee COS pricer for the same problem. Compare its accuracy and runtime against your Carr–Madan implementation across a range of moneyness and maturities.
3. By hand, derive the forward-mode AD for the Black–Scholes call price and verify that the resulting derivatives match the analytical Greeks. Document the dual-number arithmetic explicitly.
4. Use JAX to compute the full Greek vector (delta, gamma, vega, theta, vanna, volga) for a portfolio of fifty Heston-priced options. Compare the runtime against finite-difference Greeks computed by re-evaluating the COS pricer with bumped inputs.
5. Vectorize your surface revaluation using JAX vmap and measure the speedup against a naive Python loop. Report the result in nanoseconds per option-Greek-evaluation.
6. Optional advanced exercise (for students with C++ experience): compile a small Black–Scholes pricer through Enzyme or Adept and benchmark its AAD performance against your JAX implementation.

Milestone Deliverable: Milestone 2 — Fast Pricing and Greeks Library

Submit a Python (and optionally C++) library that prices European options under Black–Scholes and Heston models via characteristic function methods, computes the full Greek vector via algorithmic differentiation, and revalues the entire calibrated Coincall surface within a strict latency budget. The deliverable must include benchmarking results, accuracy validation against an independent Monte Carlo reference, and unit tests.

Datasets provided by Coincall:

Dataset	Contents and Coverage
coincall_btc_options_chain_snapshots_2025Q4.parquet	Five-minute-frequency snapshots of the BTC options chain for October–December 2025. Each snapshot includes all listed expiries with bid, ask, mid, mark, and implied volatility for every strike. Columns: timestamp, expiry, strike, type, bid, ask, mark, mark_iv, underlying_price.

coincall_calibrated_svi_surfaces_2025Q4.parquet	Pre-calibrated SVI parameters at each five-minute snapshot, provided as a baseline. Students will use these as the input to the fast revaluation pipeline.
coincall_reference_prices_2025Q4.parquet	Reference theoretical prices computed by Coincall's internal pricing engine for a representative subset of contracts. Used for accuracy validation.

Acceptance criteria:

- Full Coincall BTC options surface (typically 300–500 contracts) revalues with full Greek vector in under 5 milliseconds on a single CPU core.
- Accuracy: maximum relative error of $1e-4$ against the reference prices.
- Library exposes a clean Python API and is importable as a package.
- Unit tests cover all pricers, all Greeks, and the calibrated surface adapter.
- Benchmarking results documented in a short technical note with latency distributions, not just means.

Week 8 — Options Market Making Theory and Dynamic Hedging

Theme: extending GLFT to a multi-Greek inventory penalty, and integrating a dynamic hedging policy.

Learning Objectives

- Set up the El Aoud–Abergel framework for market making on options, in which inventory is a vector of Greeks.
- Derive the multi-Greek HJB system and identify the structure of the inventory penalty as a quadratic form.
- Understand the mechanics and economics of delta and gamma hedging, and quantify the P&L attribution of a hedged options book.
- Integrate a dynamic hedging policy with the quoting engine and identify the interaction effects.

Lecture 1 — Multi-Dimensional Inventory and the Multi-Greek HJB

The lecture states the conceptual shift required to extend market making to options. A single option position is simultaneously a position in the underlying (delta), in convexity (gamma), in volatility (vega), and in time decay (theta). The inventory state is therefore a vector, and the appropriate notion of inventory risk is a quadratic form on that vector. Following El Aoud and Abergel, the lecture derives the multi-Greek HJB and applies the now-familiar separation ansatz to reduce it to a PDE on the inventory vector. The reduction to GLFT in the one-dimensional limit is verified.

- Greek aggregation across a portfolio of options.
- The inventory vector $q = (\Delta, \Gamma, v, \dots)$ and its dynamics under quoting.
- Inventory penalty as a quadratic form: economic interpretation of the penalty matrix.
- HJB equation in the multi-Greek setting and the separation ansatz.
- Reduction to a PDE on the inventory vector; closed-form quotes under quadratic penalty.
- Truncation: which Greeks enter the optimization, which are managed by external hedging.
- Verification of the reduction to GLFT in the one-dimensional case.

Lecture 2 — Dynamic Hedging and Integration with Quoting

The market making engine produces quotes; a separate or integrated hedging policy manages the resulting Greek exposures. The lecture covers the canonical practitioner perspective on delta hedging and gamma scalping, decomposes the daily P&L of a hedged option book into its constituent streams, and then covers the two architectures for integrating hedging with quoting: external hedging by a separate algorithm that runs at lower frequency, and integrated hedging in which the quoting engine itself takes hedge orders into account. Coincall-specific issues are discussed, especially the use of the perpetual futures contract as the delta-hedging instrument.

- Continuous and discrete delta hedging; the gamma–theta–realized-variance identity.
- Gamma scalping mechanics and the P&L distribution of a long-gamma position.
- Vega hedging through other options; basis risk and skew risk.
- External hedging policies: threshold-triggered rebalancing.
- Integrated hedging: hedge orders as part of the quoting engine's action space.
- Cost-aware hedging: balancing risk reduction against transaction costs.

- Perpetual futures as the delta-hedging instrument; funding rate considerations.

Recitation

Hybrid workshop. The first half is a derivation exercise on the multi-Greek HJB, performed in pairs with the mentor circulating. The second half is a simulation exercise: students run a hedged-book simulation under various hedge policies and compare the resulting P&L distributions. The recitation closes with a discussion of which hedge policy will be used in the week 9 milestone backtest.

Required Reading

- El Aoud and Abergel (2015), A Stochastic Control Approach to Option Market Making.
- Taleb (1997), Dynamic Hedging, selected chapters on Greek aggregation, gamma scalping, and vega risk.
- Cartea, Jaikumar, and Penalva (2015), Chapter 11.

Problem Set 8

1. Derive the inventory dynamics for a market maker quoting on a portfolio of options, with inventory vector $q \in \mathbb{R}^k$ representing the aggregated Greeks. Identify the precise structure of the jumps when a fill occurs.
2. Set up the HJB equation for the El Aoud–Abergel problem and apply the separation ansatz. Reduce the equation to a PDE on the inventory vector, and show that it reduces to the GLFT HJB when only one Greek is tracked and one option is traded.
3. Choose a quadratic inventory penalty matrix that reflects the empirical covariances of delta, gamma, and vega exposures in a representative crypto options portfolio. Justify each entry from one week of Coincall data.
4. Derive the P&L of a continuously delta-hedged short call position under Black–Scholes dynamics. Show that the P&L equals the integral of half-gamma times the difference between realized and implied variance.
5. Implement a delta-hedging policy that triggers a hedge order when $|\Delta|$ exceeds 0.1 BTC of underlying exposure, and apply it to one week of Coincall data. Report the resulting hedging cost and risk reduction, and combine the hedger with your GLFT quoting engine from week 5 to observe the interaction.
6. Discuss three modeling assumptions in El Aoud and Abergel (2015) that are reasonable for equity options but less appropriate for crypto options, and propose a modification for each.

Week 9 — Backtesting and Risk Analysis

Theme: assembling the full system and validating it on historical data.

Learning Objectives

- Implement an event-driven backtest harness suitable for options market making.
- Produce a full P&L attribution that decomposes daily P&L into spread capture, inventory drift, gamma scalping, and hedging costs.
- Analyze the resulting risk metrics and identify regimes in which the model performs well or poorly.

Lecture 1 — Backtesting Architecture for Options

Backtesting an options market maker is substantially harder than backtesting a directional strategy. The lecture covers the architectural choices: an event-driven simulator that replays order book events, a fill simulator that probabilistically determines which of the market maker's quotes are filled, and a state machine that tracks the full inventory and Greek state. Common pitfalls are emphasized.

- Event-driven simulation: clock, event queue, deterministic replay.
- Fill simulator: probabilistic fills conditional on quote distance and observed order flow.
- State tracking: cash, inventory by contract, aggregated Greeks, hedge position.
- Pitfalls: look-ahead bias, unrealistic fill rates, latency assumptions, contract expiry handling.
- Stress tests: known regime shifts in the historical data.

Lecture 2 — Risk Metrics and P&L Attribution

The lecture covers the standard suite of risk and performance metrics, but emphasizes attribution. A market maker's daily P&L is the sum of several conceptually distinct streams, and isolating each is essential for understanding what the strategy is actually doing. The lecture also covers regime analysis: identifying conditions under which the strategy outperforms or underperforms.

- Sharpe ratio, Sortino ratio, maximum drawdown, Calmar ratio.
- P&L attribution: spread P&L, inventory P&L, gamma P&L, vega P&L, hedging P&L.
- Regime analysis: high-volatility versus low-volatility days.
- Greek-exposure time series and stationarity analysis.
- Capacity analysis: how P&L scales with position limits.

Recitation

Code review of the full backtest. Each student presents one regime (one week of the three-month window) and walks through the P&L attribution. The mentor probes the implementation for the pitfalls discussed in the lecture.

Required Reading

- Cartea, Jaikumar, and Penalva (2015), Chapters 10–11 (re-read).
- Bailey and López de Prado (2014), The Deflated Sharpe Ratio.
- López de Prado (2018), Advances in Financial Machine Learning, Chapters 11–13.

Problem Set 9

1. Implement an event-driven backtest harness that consumes Coincall order book and trade events and produces a deterministic replay. Validate it by running an identical configuration twice and confirming bitwise-identical results.
2. Implement a fill simulator that models the probability of fill on the market maker's quotes as a function of quote distance from mid and the observed depth of opposite-side flow. Calibrate it from historical data and validate the calibration on a holdout period.
3. Run a full one-month backtest of your multi-Greek market making engine on Coincall BTC options. Report mean and standard deviation of daily P&L, Sharpe ratio, maximum drawdown, and time series of aggregated Greeks.
4. Decompose the daily P&L into the four attribution streams discussed in lecture (spread, inventory, gamma, hedging) and produce a stacked-area plot of the cumulative attribution over the backtest period.
5. Identify the day in the three-month window with the worst drawdown and produce a forensic analysis of that day: what was the market environment, which positions drove the loss, and what (if anything) could have been done differently.

Milestone Deliverable: Milestone 3 — Full Options Market Making Backtest

Submit a complete backtest of the multi-Greek options market making engine on three months of Coincall BTC and ETH options data, with full P&L attribution, risk metrics, and Greek-exposure time series. The deliverable must include reproducible code, a configuration file documenting all hyperparameters, and a technical report that analyzes the results and identifies failure modes.

Datasets provided by Coincall:

Dataset	Contents and Coverage
coincall_btc_options_l2_2025Q4.parquet	BTC options L2 order book snapshots at 1-second resolution for October–December 2025. Five levels each side. All actively quoted contracts.
coincall_btc_options_trades_2025Q4.parquet	BTC options trade tape for the same period. Columns: timestamp, contract_id, price, size, aggressor_side, mark_iv_at_trade.
coincall_eth_options_l2_2025Q4.parquet	ETH options L2 order book snapshots at 1-second resolution for the same period.
coincall_eth_options_trades_2025Q4.parquet	ETH options trade tape for the same period.
coincall_btc_eth_perp_l2_2025Q4.parquet	BTC and ETH perpetual futures L2 snapshots at 100ms resolution. Used as the delta-hedging instrument.
coincall_funding_rates_2025Q4.parquet	Hourly perpetual futures funding rates for BTC and ETH. Required for accurate hedging cost accounting.

Acceptance criteria:

- Backtest is fully reproducible from the provided configuration file.
- P&L attribution sums to total P&L within a tolerance of 1 basis point.
- Sharpe ratio is reported with confidence interval estimated by block bootstrap.
- Greek exposure time series remain bounded within the configured position limits at all times.
- Technical report is at most twenty pages excluding appendices, and includes a forensic analysis of at least one drawdown event.

Week 10 — Extensions, Final Report, and Presentations

Theme: pursuing one novel direction and communicating results.

Learning Objectives

- Select and complete one extension of the baseline model from a menu of approved options.
- Write a final research report at the level expected of a quantitative research interview or working paper.
- Present the work clearly to a mixed academic and industry audience.

Lecture 1 — Extension Topics

The lecture surveys the menu of approved extension topics and outlines what a credible week-long investigation of each would look like. Students choose by end of day one. The topics are non-trivial but tractable within the time budget given the infrastructure built over the preceding nine weeks, and they connect directly to the GLFT-variants survey from week 5.

- Skew trading: quoting asymmetrically across moneyness based on signals derived from the calibrated SVI parameters.
- Term-structure arbitrage detection: identifying when the front and back of the term structure imply inconsistent variance forecasts.
- Machine-learned order flow prediction: training a model to predict short-horizon order arrivals as an input to the quoting engine.
- Cross-exchange market making: extending the engine to quote simultaneously on Coincall and at least one other venue.
- Robust optimization: replacing the point-estimate inventory penalty with a worst-case penalty over a confidence set.
- Multi-asset extension: extending the engine to quote BTC and ETH options simultaneously with a correlated covariance matrix.

Lecture 2 — Research Communication

The lecture covers the structure and style of a quantitative research report: how to motivate a problem, present methodology, report results, and acknowledge limitations. Examples of well-written papers in the field are discussed. The lecture closes with guidance for the oral presentation.

- Structure of a quantitative research paper: abstract, introduction, methodology, results, discussion, limitations.
- Writing for an industry audience: avoiding both excessive formalism and excessive informality.
- Presenting empirical results: figures, tables, and the use of confidence intervals.
- Acknowledging limitations: a sign of strength, not weakness.
- Oral presentation: structure, pacing, and the use of slides.

Recitation

Final dry-run presentations. Each student gives a fifteen-minute presentation to the cohort, and the cohort provides feedback. The mentor focuses on questions that an industry audience would ask.

Required Reading

- El Aoud and Abergel (2015), re-read with the extension topic in mind.
- Guéant (2016), the chapter most relevant to the chosen extension.
- One additional paper relevant to the chosen extension, identified by the student in consultation with the mentor.

Problem Set 10

1. Submit a one-page proposal for your chosen extension by end of day one of week 10, including problem statement, methodology, expected results, and risk factors.
2. Implement the extension on top of the infrastructure from weeks 5, 7, and 9.
3. Quantify the improvement (or lack thereof) over the baseline established in week 9 using the same risk and P&L metrics.
4. Write a final technical report not exceeding thirty pages, structured following the lecture's guidance, and incorporating all previous milestone work.
5. Prepare and deliver a twenty-minute final presentation to the cohort and invited industry guests.

3. Comprehensive Reference List

The references below are organized by the role they play in the program. Primary references are required reading; secondary references provide context and depth; supplementary references support specific weeks or extension topics.

3.1 Primary References

- Avellaneda, M., and Stoikov, S. (2008). High-frequency trading in a limit order book. *Quantitative Finance*, 8(3), 217–224.
- Guéant, O., Lehalle, C.-A., and Fernandez-Tapia, J. (2013). Dealing with the inventory risk: a solution to the market making problem. *Mathematics and Financial Economics*, 7(4), 477–507.
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- El Aoud, S., and Abergel, F. (2015). A stochastic control approach to option market making. *Market Microstructure and Liquidity*, 1(1), 1550006.
- Cartea, Á., Jaikumar, S., and Penalva, J. (2015). *Algorithmic and High-Frequency Trading*. Cambridge University Press.

3.2 Volatility Surface and Options Pricing

- Gatheral, J. (2006). *The Volatility Surface: A Practitioner's Guide*. Wiley Finance.
- Gatheral, J., and Jacquier, A. (2014). Arbitrage-free SVI volatility surfaces. *Quantitative Finance*, 14(1), 59–71.
- Hagan, P. S., Kumar, D., Lesniewski, A. S., and Woodward, D. E. (2002). Managing smile risk. *Wilmott Magazine*, September, 84–108.
- Hull, J. C. (2021). *Options, Futures, and Other Derivatives* (11th ed.). Pearson.
- Taleb, N. N. (1997). *Dynamic Hedging: Managing Vanilla and Exotic Options*. Wiley.

3.3 Fast Computation Methods

- Carr, P., and Madan, D. (1999). Option valuation using the fast Fourier transform. *Journal of Computational Finance*, 2(4), 61–73.
- Fang, F., and Oosterlee, C. W. (2008). A novel pricing method for European options based on Fourier-cosine series expansions. *SIAM Journal on Scientific Computing*, 31(2), 826–848.
- Giles, M., and Glasserman, P. (2006). Smoking adjoints: fast Monte Carlo Greeks. *Risk Magazine*, January, 88–92.
- Capriotti, L. (2011). Fast Greeks by algorithmic differentiation. *Journal of Computational Finance*, 14(3), 3–35.
- Glasserman, P. (2003). *Monte Carlo Methods in Financial Engineering*. Springer.

3.4 Stochastic Optimal Control

Pham, H. (2009). *Continuous-time Stochastic Control and Optimization with Financial Applications*. Springer.

Fleming, W. H., and Soner, H. M. (2006). *Controlled Markov Processes and Viscosity Solutions* (2nd ed.). Springer.

Øksendal, B. (2003). *Stochastic Differential Equations: An Introduction with Applications* (6th ed.). Springer.

3.5 Market Microstructure

Roll, R. (1984). A simple implicit measure of the effective bid-ask spread in an efficient market. *Journal of Finance*, 39(4), 1127–1139.

Glosten, L. R., and Milgrom, P. R. (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1), 71–100.

Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica*, 53(6), 1315–1335.

O'Hara, M. (1995). *Market Microstructure Theory*. Blackwell.

3.6 Supplementary

Bailey, D. H., and López de Prado, M. (2014). The deflated Sharpe ratio: correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5), 94–107.

López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.

Almgren, R., and Chriss, N. (2001). Optimal execution of portfolio transactions. *Journal of Risk*, 3(2), 5–40.

Appendix A. Coincall Dataset Catalog

The following datasets are provided by Coincall to the program. All datasets are delivered as Parquet files on a secure object store accessible to enrolled students under a non-disclosure agreement. The naming convention is consistent across datasets: `coincall_{instrument}_{data_type}_{period}.parquet`.

A.1 Order Book Data

Order book data is provided as L2 snapshots, with the top ten price levels on each side aggregated by quantity. Snapshots are taken at fixed intervals (1 second for options, 100 milliseconds for perpetual futures). Each snapshot row contains the bid and ask prices and sizes at each level, a sequence number that strictly increases within the file, and an exchange timestamp.

The L2 representation is preferred over L3 for this program for two reasons: it is sufficient for both calibrating intensity functions and constructing the simulation environment, and it is several orders of magnitude smaller, which keeps the data engineering tractable for a ten-week program.

A.2 Trade Data

Trade data is provided as a tape of all executed trades within the period. Each trade includes the contract identifier, price, size, the aggressor side (whether the trade was initiated by a buyer or seller), and a Coincall trade identifier that can be cross-referenced with the order book data to identify which resting order was filled.

For options, each trade additionally includes the mark implied volatility at the time of the trade. This is useful for the volatility surface calibration in week 6 and for the fill simulator's calibration in week 9.

A.3 Surface Data

Calibrated SVI parameters are provided at five-minute frequency as a baseline for the fast revaluation pipeline in week 7. Students are expected to also produce their own calibrations as part of week 6, but the provided baseline is useful for sanity-checking and for downstream pipelines.

A.4 Auxiliary Data

Funding rate data for the perpetual futures contracts is provided at hourly frequency. This is required for accurate accounting of the hedging cost in week 9. Additionally, a snapshot of the exchange's contract specifications (tick size, lot size, fee schedule, position limits) is provided as a static file.

A.5 Access and Use

Students will receive credentials to a secure S3-compatible endpoint at the start of the program. Data may be downloaded locally for analysis but may not be redistributed or used for any purpose other than the program. The full terms are specified in the data use agreement that each student signs at enrollment.

Coincall retains the right to update the dataset catalog during the program in response to material market events or changes in available data. Any such updates will be communicated to students and the program coordinator with at least one week of notice.